

CPPDescent

713bf61 (with uncommitted changes)

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Chapter 1

CPP-Descent

Part of the “Software Development for Computing Systems” course of the Department of Informatics and Telecommunications, UoA for the first semester of the academic year 2023-2024.

A library that implements the KNN-Graph creation and search algorithms described in [1], [2].

1.1 Dependencies

For local development, you will need Make and CMake. To install them on an apt-based linux distro, run the following commands:

```
$ sudo apt update && sudo apt upgrade
$ sudo apt install make
$ sudo apt install cmake
```

1.1.1 Minor dependencies

- The project uses [Google Test](#) for its testing needs. Gtest is used as a git submodule of this repo, that is downloaded by CMake the first time you clone. That means that nothing extra needs to be installed to run the tests.
- If you would like to generate the documentation, you will need [doxygen](#) installed.

1.2 Code Structure

In the [wiki](#), you can find documentation generated by [Doxygen](#). It is also available in [pdf form](#).

1.2.1 Code Coverage

This project uses [LCOV](#) for its code coverage needs. The results are uploaded upon commit [here](#).

1.3 Contributors

- [Konstantinos Chousos](#) (1115202000215)
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1.4 Bibliography

[1] W. Dong, C. Moses, and K. Li, “Efficient k-nearest neighbor graph construction for generic similarity measures,” in *Proceedings of the 20th international conference on World wide web*, in WWW '11. New York, NY, USA: Association for Computing Machinery, Mar. 2011, pp. 577–586. doi: [10.1145/1963405.1963487](#).

[2] “How PyNNDescent works — pynn descent 0.5.0 documentation.” Accessed: Oct. 10, 2023. [Online]. Available: https://pynn descent.readthedocs.io/en/latest/how_pynn descent_works.html

Chapter 2

Class Index

2.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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Chapter 3

File Index

3.1 File List

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Chapter 4

Class Documentation

4.1 cppdescent Class Reference

A sample function for the cppdescent class, used to test the correctness of the CMake files.

```
#include <cppdescent.hpp>
```

Public Member Functions

- int [hello](#) (void)
Prints "Hello World".

4.1.1 Detailed Description

A sample function for the cppdescent class, used to test the correctness of the CMake files.

4.1.2 Member Function Documentation

4.1.2.1 hello()

```
int cppdescent::hello (  
    void )
```

Prints "Hello World".

Returns

int exit code

The documentation for this class was generated from the following files:

- [cppdescent.hpp](#)
- [cppdescent.cpp](#)

4.2 List Class Reference

ADT [List](#).

```
#include <ADTList.hpp>
```

Public Member Functions

- [List](#) (DestroyFunc value=nullptr)
Construct a new [List](#) object.
- [~List](#) ()
Destroy the [List](#) object.
- void [insertNext](#) ([ListNode](#) *node, Pointer value)
Adds a node after the given node.
- void [removeNext](#) ([ListNode](#) *node)
Removes the node after the given node.
- Pointer [find](#) (Pointer value, CompareFunc compare)
Finds the first value equal to the value parameter.
- DestroyFunc [setDestroyValue](#) (DestroyFunc value)
Set the Destroy Value object to a new value.
- int [getSize](#) ()
Get the size of the [List](#).
- int [increaseSize](#) ()
Increase the list size.
- int [decreaseSize](#) ()
Decrease the list size.
- [ListNode](#) * [getHead](#) ()
Getter for the head of the [List](#).
- [ListNode](#) * [getTail](#) ()
Getter for the tail of the [List](#).
- [ListNode](#) * [next](#) ([ListNode](#) *node)
Returns the node after the given one.
- Pointer [nodeValue](#) ([ListNode](#) *node)
Returns the value of the node.
- [ListNode](#) * [findNode](#) (Pointer value, CompareFunc compare)
Finds the first node that has value equal to the value parameter.

4.2.1 Detailed Description

ADT [List](#).

An Abstract Data Type [List](#) with linear access to data, and addition and removal to/from arbitrary positions.

4.2.2 Constructor & Destructor Documentation

4.2.2.1 List()

```
List::List (
    DestroyFunc value = nullptr )
```

Construct a new [List](#) object.

Parameters

<i>destroyValue</i>	If <i>destroyValue</i> != nullptr, the <i>destroyValue</i> function will be called each time a node is removed.
---------------------	---

head is a virtual node that is always present in the list.

Parameters

<i>destroyValue</i>	The destroy function for the values of the list. Defaults to nullptr.
---------------------	---

4.2.2.2 ~List()

```
List::~~List ( )
```

Destroy the [List](#) object.

If the *destroyValue* function is not nullptr, it is called for every value.

4.2.3 Member Function Documentation**4.2.3.1 decreaseSize()**

```
int List::decreaseSize ( )
```

Decrease the list size.

Returns

int The updated size.

4.2.3.2 find()

```
Pointer List::find (
    Pointer value,
    CompareFunc compare )
```

Finds the first value equal to the value parameter.

Find the first value equal to the value parameter.

Parameters

<i>value</i>	The value based on which the search is done.
<i>compare</i>	A pointer to the function that will compare the node values.

Returns

Pointer A pointer to the resulting node.

Parameters

<i>value</i>	The value to search for.
<i>compare</i>	A pointer to the compare function.

Returns

Pointer A pointer to found value.

4.2.3.3 findNode()

```
ListNode * List::findNode (
    Pointer value,
    CompareFunc compare )
```

Finds the first node that has value equal to the value parameter.

Find the first node with value equal to the value parameter.

Parameters

<i>value</i>	The value to search.
<i>compare</i>	The compare function to use.

Returns

ListNode* A pointer to the resulting node.

Parameters

<i>value</i>	The value to search for.
<i>compare</i>	A pointer to the compare function.

Returns

ListNode* A pointer to the found node.

4.2.3.4 getHead()

```
ListNode * List::getHead ( )
```

Getter for the head of the [List](#).

Get the first element of the [List](#).

Returns

ListNode* The head.

Returns

ListNode* Pointer to the first element.

4.2.3.5 getSize()

```
int List::getSize ( )
```

Get the size of the [List](#).

Get the size of the list.

The size is updated upon entry/removal of nodes, so the retrieval of the size is O(1).

Returns

int

Returns

int The size.

4.2.3.6 getTail()

```
ListNode * List::getTail ( )
```

Getter for the tail of the [List](#).

Get the last element of the list.

Returns

ListNode* The tail.

If the list is empty, the virtual head is the tail, so we return nullptr.

Returns

ListNode* Pointer to the last element.

4.2.3.7 increaseSize()

```
int List::increaseSize ( )
```

Increase the list size.

Returns

int The updated size.

4.2.3.8 insertNext()

```
void List::insertNext (
    ListNode * node,
    Pointer value )
```

Adds a node *after* the given node.

Insert a new element after the node.

Parameters

<i>node</i>	The node that the new node will be placed after.
<i>value</i>	The value of the new node.

Parameters

<i>node</i>	The node after which the element is inserted.
<i>value</i>	The value of the new element.

4.2.3.9 next()

```
ListNode * List::next (
    ListNode * node )
```

Returns the node after the given one.

Get the next node.

Parameters

<i>node</i>	
-------------	--

Returns

ListNode* A pointer to the next node.

Parameters

<i>node</i>	
-------------	--

Returns

ListNode*

4.2.3.10 nodeValue()

```
Pointer List::nodeValue (  
    ListNode * node )
```

Returns the value of the node.

Get the node's value.

Parameters

<i>node</i>	
-------------	--

Returns

Pointer A generic pointer to the value.

Parameters

<i>node</i>	
-------------	--

Returns

Pointer

4.2.3.11 removeNext()

```
void List::removeNext (  
    ListNode * node )
```

Removes the node *after* the given node.

Removes the node after the given node.

Parameters

<i>node</i>	The node before the node to be removed.
-------------	---

Parameters

<i>node</i>	The node after which the element is removed.
-------------	--

4.2.3.12 setDestroyValue()

```
DestroyFunc List::setDestroyValue (
    DestroyFunc value )
```

Set the Destroy Value object to a new value.

Set the destroy value function.

Parameters

<i>value</i>	A pointer to the new destroy function.
--------------	--

Returns

DestroyFunc A pointer to the old destroy function.

Parameters

<i>value</i>	The new destroy function.
--------------	---------------------------

Returns

DestroyFunc The old destroy function.

The documentation for this class was generated from the following files:

- ADTList.hpp
- [ADTList.cpp](#)

4.3 ListNode Class Reference

Node of a list.

```
#include <ADTList.hpp>
```

Public Member Functions

- [ListNode](#) ()
Construct a new [List](#) Node object that is empty.
- [ListNode](#) (Pointer value)
Construct a new [List](#) Node object with value 'value'.
- void [setNext](#) ([ListNode](#) *next)
Setter for the next node.
- [ListNode](#) * [getNext](#) ()
Getter for the next node.
- Pointer [getValue](#) ()
Getter for the value of the node.

4.3.1 Detailed Description

Node of a list.

A list node is represented by its value. It also holds a pointer to the next node in the list.

4.3.2 Constructor & Destructor Documentation

4.3.2.1 [ListNode\(\)](#) [1/2]

```
ListNode::ListNode ( ) [inline]
```

Construct a new [List](#) Node object that is empty.

4.3.2.2 [ListNode\(\)](#) [2/2]

```
ListNode::ListNode (
    Pointer value ) [inline]
```

Construct a new [List](#) Node object with value 'value'.

Parameters

<i>value</i>	a generic pointer to the value of the node.
--------------	---

4.3.3 Member Function Documentation

4.3.3.1 getNext()

```
ListNode * ListNode::getNext ( )
```

Getter for the next node.

Returns

ListNode* The pointer to the next node.

4.3.3.2 getValue()

```
Pointer ListNode::getValue ( )
```

Getter for the value of the node.

Returns

Pointer A generic pointer to the value.

4.3.3.3 setNext()

```
void ListNode::setNext (
    ListNode * next )
```

Setter for the next node.

Parameters

<i>next</i>	The pointer to the next node.
-------------	-------------------------------

The documentation for this class was generated from the following files:

- ADTList.hpp
- [listNode.cpp](#)

4.4 PQueue Class Reference

ADT Priority Queue.

```
#include <ADTPQueue.hpp>
```

Public Member Functions

- [PQueue](#) (CompareFunc compare, DestroyFunc destroyValue, [Vector](#) *values)
Construct a new [PQueue](#) object.
- [~PQueue](#) ()
Destroy the [PQueue](#) object.
- int [getSize](#) ()
Get the size of the priority queue.
- Pointer [getMax](#) ()
Get the max element of the queue.
- void [insert](#) (Pointer value)
Insert a new element to the queue.
- void [removeMax](#) ()
Remove the max of the queue.
- DestroyFunc [setDestroyValue](#) (DestroyFunc destroyValue)
Set the Destroy Value object.
- Pointer **nodeValue** (int nodeId)
- void **nodeSwap** (int nodeId1, int nodeId2)
- void **bubbleUp** (int nodeId)
- void **bubbleDown** (int nodeId)
- void **naiveHeapify** ([Vector](#) *values)

4.4.1 Detailed Description

ADT Priority Queue.

4.4.2 Constructor & Destructor Documentation

4.4.2.1 PQueue()

```
PQueue::PQueue (
    CompareFunc compare,
    DestroyFunc destroyValue,
    Vector * values )
```

Construct a new [PQueue](#) object.

Uses the `compare` function to compare its elements.

If `destroyValue != nullptr`, the `destroyValue` function is called upon removal of an element.

If `values != nullptr`, then the priority queue will be initialized with the values of the vector `values`.

Parameters

<i>compare</i>	
<i>destroyValue</i>	
<i>values</i>	

4.4.2.2 ~PQueue()

```
PQueue::~~PQueue ( )
```

Destroy the [PQueue](#) object.

4.4.3 Member Function Documentation

4.4.3.1 getMax()

```
Pointer PQueue::getMax ( )
```

Get the max element of the queue.

Returns

Pointer A generic pointer to the max element.

4.4.3.2 getSize()

```
int PQueue::getSize ( )
```

Get the size of the priority queue.

Returns

int

4.4.3.3 insert()

```
void PQueue::insert (
    Pointer value )
```

Insert a new element to the queue.

Parameters

<i>value</i>	The value of the new element to be inserted.
--------------	--

4.4.3.4 removeMax()

```
void PQueue::removeMax ( )
```

Remove the max of the queue.

4.4.3.5 setDestroyValue()

```
DestroyFunc PQueue::setDestroyValue (
    DestroyFunc destroyValue )
```

Set the Destroy Value object.

Parameters

<i>destroyValue</i>	
---------------------	--

Returns

DestroyFunc

The documentation for this class was generated from the following files:

- [ADTPQueue.hpp](#)
- [ADTPQueue.cpp](#)

4.5 Vector Class Reference

ADT [Vector](#).

```
#include <ADTVector.hpp>
```

Public Member Functions

- [Vector](#) (int size, DestroyFunc destroyValue)
Construct a new [Vector](#) object.
- [~Vector](#) ()
Destroy the [Vector](#) object.
- int [getSize](#) ()
Get the current size of the [Vector](#).
- void [insertLast](#) (Pointer value)
Inserts a new element at the end of the vector.

- `int removeLast ()`
Removes the last element of the vector.
- `Pointer getAt (int pos)`
Get the value at the specified position of the vector.
- `int setAt (int pos, Pointer value)`
Set the value at the specified position of the vector.
- `Pointer find (Pointer value, CompareFunc compare)`
Find the first element with value equal to value.
- `DestroyFunc setDestroyValue (DestroyFunc destroyValue)`
Set the Destroy Value.
- `vectorNode * first ()`
Get the first node of the vector.
- `vectorNode * last ()`
Get the last node of the vector.
- `vectorNode * next (vectorNode *node)`
Get the next node of the vector, after the one given.
- `vectorNode * previous (vectorNode *node)`
Get the previous node of the vector, before the one given.
- `Pointer nodeValue (vectorNode *node)`
Get the value of the node.
- `vectorNode * findNode (Pointer value, CompareFunc compare)`
Find the first node with value equal to value.

4.5.1 Detailed Description

ADT [Vector](#).

An ADT [Vector](#) using a dynamic array for smart resizing.

4.5.2 Constructor & Destructor Documentation

4.5.2.1 Vector()

```
Vector::Vector (
    int size,
    DestroyFunc destroyValue )
```

Construct a new [Vector](#) object.

The newly created [Vector](#) will be of size `size` and its elements will be initialized as `nullptr`.

If `destroyValue` is not `nullptr`, it will be called on each element when it is removed from the [Vector](#).

Parameters

<code>size</code>	The initial size of the Vector .
<code>destroyValue</code>	The function to be called on each element when it is removed from the Vector .

4.5.2.2 ~Vector()

```
Vector::~~Vector ( )
```

Destroy the [Vector](#) object.

Deletes all allocated memory of the [Vector](#).

4.5.3 Member Function Documentation

4.5.3.1 find()

```
Pointer Vector::find (
    Pointer value,
    CompareFunc compare )
```

Find the first element with value equal to value.

Parameters

<i>value</i>	The value to look for.
<i>compare</i>	The function to be used for comparison.

Returns

Pointer A pointer to the found value.

4.5.3.2 findNode()

```
vectorNode * Vector::findNode (
    Pointer value,
    CompareFunc compare )
```

Find the first node with value equal to value.

Parameters

<i>value</i>	The value to look for.
<i>compare</i>	The function to be used for comparison.

Returns

[vectorNode](#) The resulting node.

4.5.3.3 first()

```
vectorNode * Vector::first ( )
```

Get the first node of the vector.

Returns

[vectorNode](#)

4.5.3.4 getAt()

```
Pointer Vector::getAt (
    int pos )
```

Get the value at the specified position of the vector.

Parameters

<i>pos</i>	The position to look for.
------------	---------------------------

Returns

Pointer A pointer to the value at the specified position.

4.5.3.5 getSize()

```
int Vector::getSize ( )
```

Get the current size of the [Vector](#).

Returns

int The size of the [Vector](#).

4.5.3.6 insertLast()

```
void Vector::insertLast (
    Pointer value )
```

Inserts a new element at the end of the vector.

Parameters

<i>value</i>	The value of the new element.
--------------	-------------------------------

If we are at the last element, we create a new array double the size.

4.5.3.7 last()

```
vectorNode * Vector::last ( )
```

Get the last node of the vector.

Returns

[vectorNode](#)

4.5.3.8 next()

```
vectorNode * Vector::next (
    vectorNode * node )
```

Get the next node of the vector, after the one given.

Parameters

<i>node</i>	The node to get the next of.
-------------	------------------------------

Returns

[vectorNode](#)

4.5.3.9 nodeValue()

```
Pointer Vector::nodeValue (
    vectorNode * node )
```

Get the value of the node.

Parameters

<i>node</i>	
-------------	--

Returns

Pointer

4.5.3.10 previous()

```
vectorNode * Vector::previous (
    vectorNode * node )
```

Get the previous node of the vector, before the one given.

Parameters

<i>node</i>	The node to get the previous of.
-------------	----------------------------------

Returns

[vectorNode](#)

4.5.3.11 removeLast()

```
int Vector::removeLast ( )
```

Removes the last element of the vector.

Returns

int -1 if the function fails, else 0.

4.5.3.12 setAt()

```
int Vector::setAt (
    int pos,
    Pointer value )
```

Set the value at the specified position of the vector.

Parameters

<i>pos</i>	The position to edit.
<i>value</i>	The new value.

Returns

int -1 if the function fails, else 0.

4.5.3.13 setDestroyValue()

```
DestroyFunc Vector::setDestroyValue (
    DestroyFunc destroyValue )
```

Set the Destroy Value.

Parameters

<i>destroyValue</i>	The new destroy value function.
---------------------	---------------------------------

Returns

DestroyFunc The old destroy value function.

The documentation for this class was generated from the following files:

- [ADTVector.hpp](#)
- [ADTVector.cpp](#)

4.6 vectorNode Class Reference

Class for the vector Node object.

```
#include <ADTVector.hpp>
```

Public Member Functions

- [vectorNode](#) ()
Construct a new vector Node object.
- **vectorNode** (Pointer value)
- Pointer [getValue](#) () const
Get the Value object.
- void [setValue](#) (Pointer value)
Set the Value object.

4.6.1 Detailed Description

Class for the vector Node object.

A vector node is described only its value.

4.6.2 Constructor & Destructor Documentation

4.6.2.1 vectorNode()

```
vectorNode::vectorNode ( ) [inline]
```

Construct a new vector Node object.

Parameters

<i>value</i>	A Pointer to the value.
--------------	-------------------------

4.6.3 Member Function Documentation

4.6.3.1 getValue()

```
Pointer vectorNode::getValue ( ) const [inline]
```

Get the Value object.

Returns

Pointer

4.6.3.2 setValue()

```
void vectorNode::setValue (
    Pointer value ) [inline]
```

Set the Value object.

Parameters

<i>value</i>	
--------------	--

The documentation for this class was generated from the following file:

- [ADTVector.hpp](#)

Chapter 5

File Documentation

5.1 ADTList.hpp

```
00001
00011 #pragma once
00012 #include "common.hpp"
00013
00014 #define LIST_BOF (ListNode*)0
00015 #define LIST_EOF (ListNode*)0
00016
00024 class ListNode {
00025 public:
00030     ListNode() : next(nullptr), value(nullptr){};
00036     ListNode(Pointer value) : next(nullptr), value(value){};
00042     void setNext(ListNode* next);
00048     ListNode* getNext();
00054     Pointer getValue();
00055
00056 private:
00057     ListNode* next;
00058     Pointer value;
00059 };
00060
00068 class List {
00069 public:
00076     List(DestroyFunc value = nullptr);
00081     ~List();
00088     void insertNext(ListNode* node, Pointer value);
00094     void removeNext(ListNode* node);
00102     Pointer find(Pointer value, CompareFunc compare);
00109     DestroyFunc setDestroyValue(DestroyFunc value);
00118     int getSize();
00119     int increaseSize();
00120     int decreaseSize();
00126     ListNode* getHead();
00132     ListNode* getTail();
00139     ListNode* next(ListNode* node);
00146     Pointer nodeValue(ListNode* node);
00154     ListNode* findNode(Pointer value, CompareFunc compare);
00155
00156 private:
00157     DestroyFunc destroyValue;
00158     int size;
00159     ListNode* head;
00160     ListNode* tail;
00161 };
```

5.2 ADTPQueue.hpp File Reference

```
#include "ADTVector.hpp"
#include "common.hpp"
```

Include dependency graph for ADTPQueue.hpp: This graph shows which files directly or indirectly include this file:

Classes

- class [PQueue](#)
ADT Priority Queue.

5.2.1 Detailed Description

Author

Konstantinos Chousos

Version

0.1

Date

2023-10-30

Copyright

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5.3 ADTPQueue.hpp

[Go to the documentation of this file.](#)

```
00001
00011 #pragma once
00012
00013 #include "ADTVector.hpp"
00014 #include "common.hpp"
00015
00020 class PQueue {
00021 public:
00037     PQueue(CompareFunc compare, DestroyFunc destroyValue, Vector* values);
00042     ~PQueue();
00048     int getSize();
00054     Pointer getMax();
00060     void insert(Pointer value);
00065     void removeMax();
00072     DestroyFunc setDestroyValue(DestroyFunc destroyValue);
00073
00074     // Helper functions
00075     // These are used because the node IDs are 1-based, where as the
00076     // vector is 0-based.
00077     Pointer nodeValue(int nodeId);
00078     void nodeSwap(int nodeId1, int nodeId2);
00079     void bubbleUp(int nodeId);
00080     void bubbleDown(int nodeId);
00081     void naiveHeapify(Vector* values);
00082
00083 private:
00084     Vector* vector;
00085     CompareFunc compare;
00086     DestroyFunc destroyValue;
00087 };
```

5.4 ADTVector.hpp File Reference

ADTVector implementation using a dynamic array.

```
#include "common.hpp"
```

Include dependency graph for ADTVector.hpp: This graph shows which files directly or indirectly include this file:

Classes

- class [vectorNode](#)
Class for the vector Node object.
- class [Vector](#)
ADT [Vector](#).

Macros

- `#define VECTOR_MIN_CAPACITY 10`
- `#define VECTOR_BOF (vectorNode\*)0`
- `#define VECTOR_EOF (vectorNode\*)0`

5.4.1 Detailed Description

ADTVector implementation using a dynamic array.

Author

Konstantinos Chousos

Version

0.1

Date

2023-10-29

Copyright

Copyright (c) 2023

5.5 ADTVector.hpp

[Go to the documentation of this file.](#)

```
00001
00011 #pragma once
00012
00013 #include "common.hpp"
00014
00015 #define VECTOR_MIN_CAPACITY 10
00016
00017 #define VECTOR_BOF (vectorNode\*)0
00018 #define VECTOR_EOF (vectorNode\*)0
00019
00026 class vectorNode {
00027     public:
00033     vectorNode(){};
00034     vectorNode(Pointer value) : value(value){};
00040     Pointer getValue() const { return value; };
00046     void setValue(Pointer value) { this->value = value; };
00047
00048     private:
00049     Pointer value;
00050 };
00051
```

```

00058 class Vector {
00059 public:
00073   Vector(int size, DestroyFunc destroyValue);
00080   ~Vector();
00086   int getSize();
00092   void insertLast(Pointer value);
00098   int removeLast();
00105   Pointer getAt(int pos);
00113   int setAt(int pos, Pointer value);
00121   Pointer find(Pointer value, CompareFunc compare);
00128   DestroyFunc setDestroyValue(DestroyFunc destroyValue);
00134   vectorNode* first();
00140   vectorNode* last();
00147   vectorNode* next(vectorNode* node);
00154   vectorNode* previous(vectorNode* node);
00161   Pointer nodeValue(vectorNode* node);
00169   vectorNode* findNode(Pointer value, CompareFunc compare);
00170
00171 private:
00172   vectorNode* array;
00173   int size;
00174   int capacity;
00175   DestroyFunc destroyValue;
00176 };

```

5.6 common.hpp

```

00001 #pragma once
00002
00003 typedef void* Pointer;
00004
00005 typedef int (*CompareFunc)(Pointer a, Pointer b);
00006
00007 typedef void (*DestroyFunc)(Pointer value);

```

5.7 cppdescent.hpp

```

00001
00006 class cppdescent {
00007 public:
00013   int hello(void);
00014 };

```

5.8 ADTList.cpp File Reference

Implementation of an ADT linked list.

```
#include "cppdescent/ADTList.hpp"
```

Include dependency graph for ADTList.cpp:

5.9 ADTPQueue.cpp File Reference

An ADT Priority Queue implemented using a heap.

```
#include "cppdescent/ADTPQueue.hpp"
#include <iostream>
```

Include dependency graph for ADTPQueue.cpp:

5.9.1 Detailed Description

An ADT Priority Queue implemented using a heap.

Author

Konstantinos Chousos

Version

0.1

Date

2023-10-30

Copyright

Copyright (c) 2023

5.10 ADTVector.cpp File Reference

Implementation of ADTVector using a dynamic array.

```
#include "cppdescent/ADTVector.hpp"  
#include "stdlib.h"  
Include dependency graph for ADTVector.cpp:
```

5.10.1 Detailed Description

Implementation of ADTVector using a dynamic array.

Author

Konstantinos Chousos

Version

0.1

Date

2023-10-30

Copyright

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5.11 cppdescent.cpp File Reference

Implementation of cppdescent library.

```
#include "cppdescent/cppdescent.hpp"
#include <iostream>
```

Include dependency graph for cppdescent.cpp:

5.11.1 Detailed Description

Implementation of cppdescent library.

Author

Konstantinos Chousos

Version

0.1

Date

2023-10-27

Copyright

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5.12 listNode.cpp File Reference

Implementation of the listNode to be used with linkedList.

```
#include "cppdescent/ADTList.hpp"
```

Include dependency graph for listNode.cpp:

5.12.1 Detailed Description

Implementation of the listNode to be used with linkedList.

Author

Konstantinos Chousos

Version

0.1

Date

2023-10-29

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